EXPERIMENT 7

Theory

Blockchain is a decentralized and secure digital ledger used to store records in a transparent and tamper-resistant manner. In **Supply Chain Management**, blockchain helps in tracking products, shipments, and transactions across different stakeholders such as suppliers, manufacturers, transporters, distributors, and buyers.

In this experiment, **MetaMask** is used as an embedded wallet to connect with the Ethereum blockchain and approve transactions. **Solidity** is used to write smart contracts, and **Remix IDE** is used to compile and deploy them.

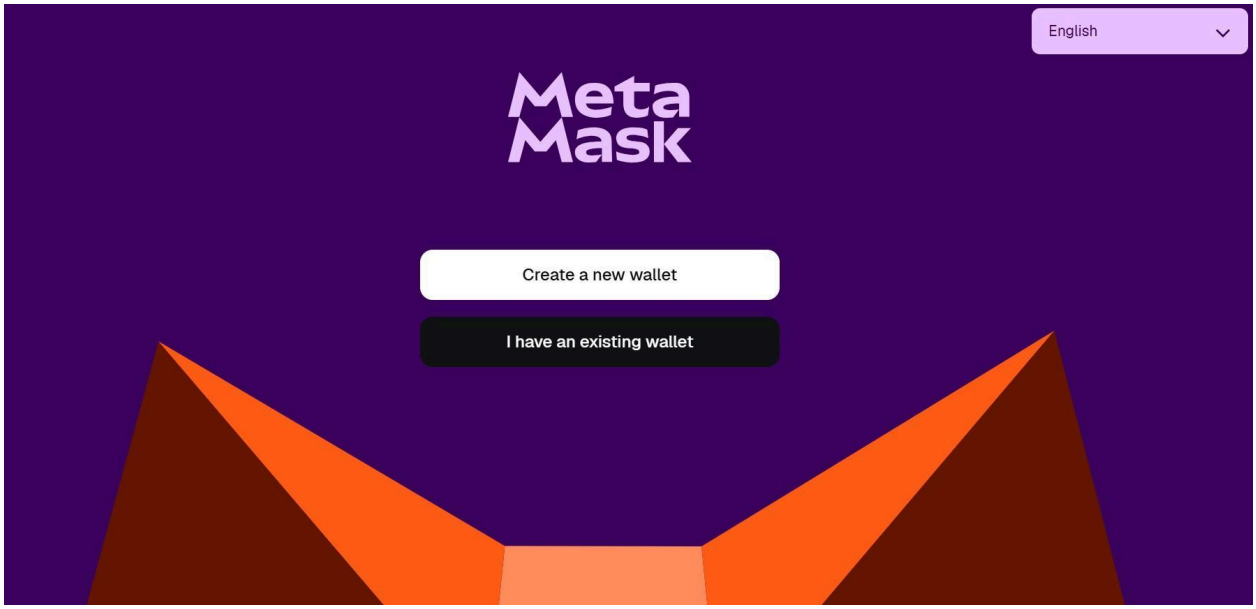
The experiment is based on three main smart contracts:

- **SupplyChainToken** – used for digital payment transactions
- **SupplyChainEscrow** – used to securely hold and release payments
- **SupplyChainRegistry** – used to store product and shipment details

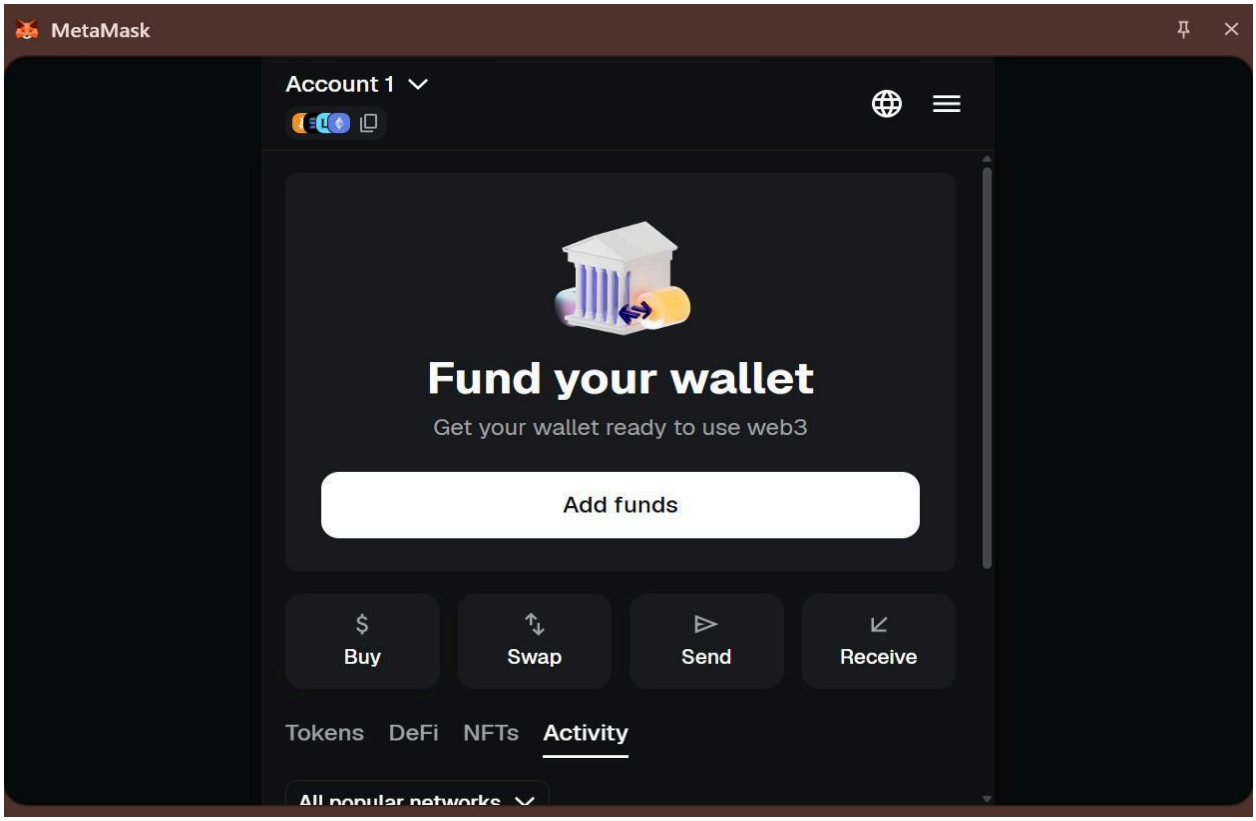
This experiment demonstrates how blockchain can improve the supply chain process by providing transparency, traceability, security, and automated transaction

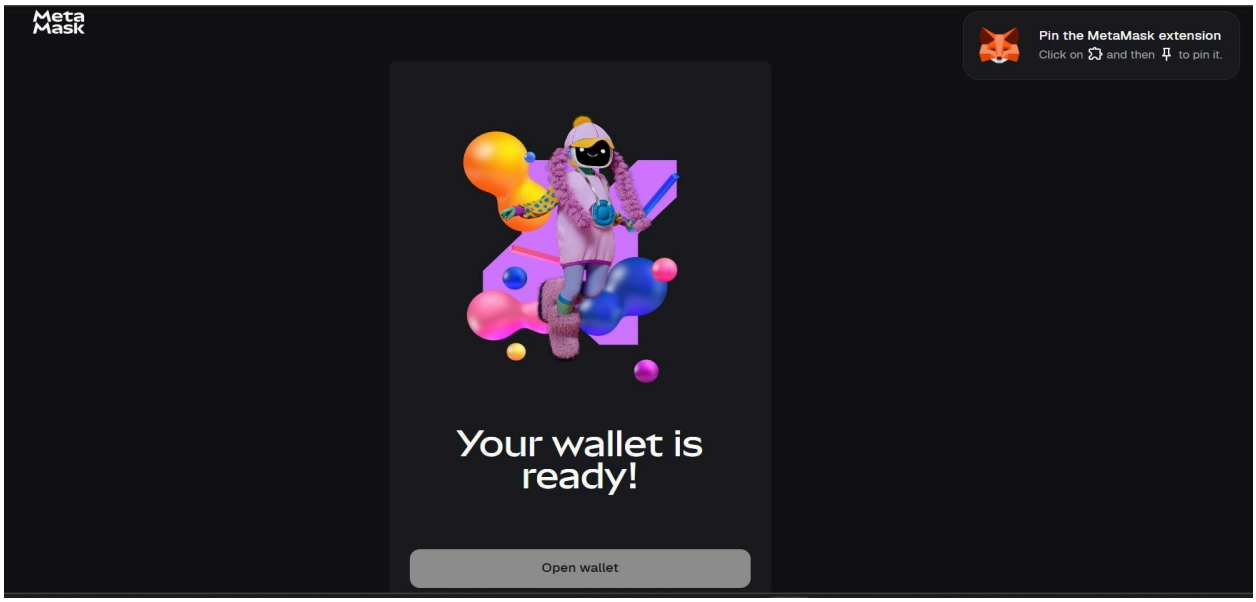
Steps:
Implementation

Step 1: Install MetaMask Install MetaMask extension and create a wallet

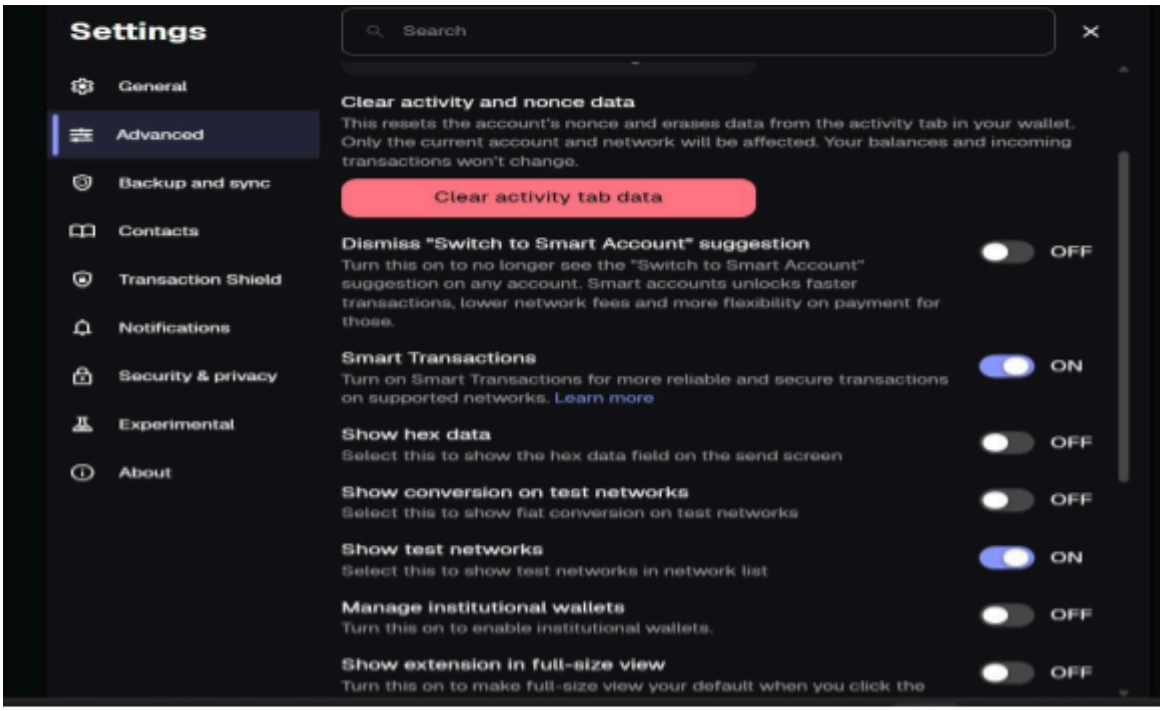


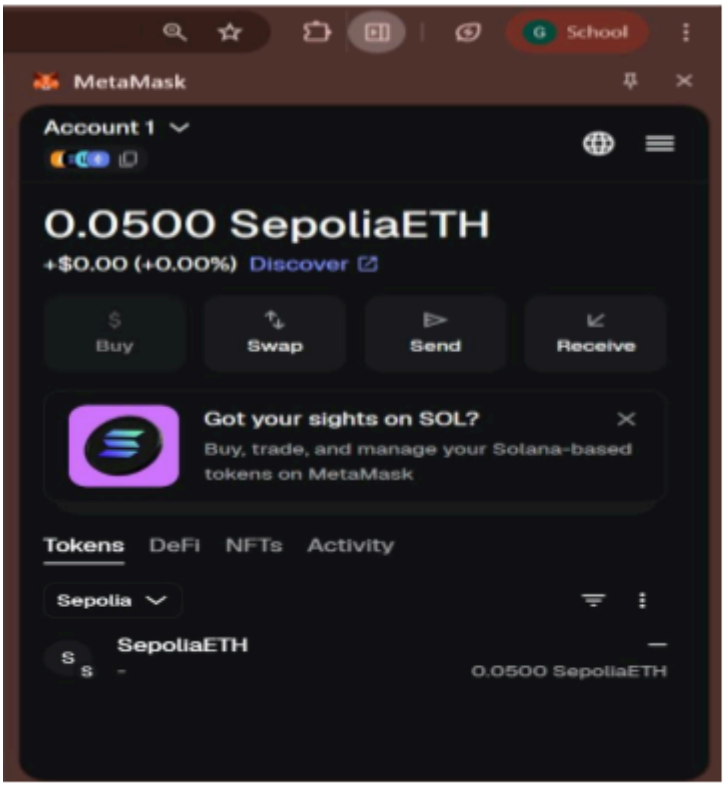
Step 2:
Add Sepolia Testnet Configure MetaMask to connect to Sepolia network.



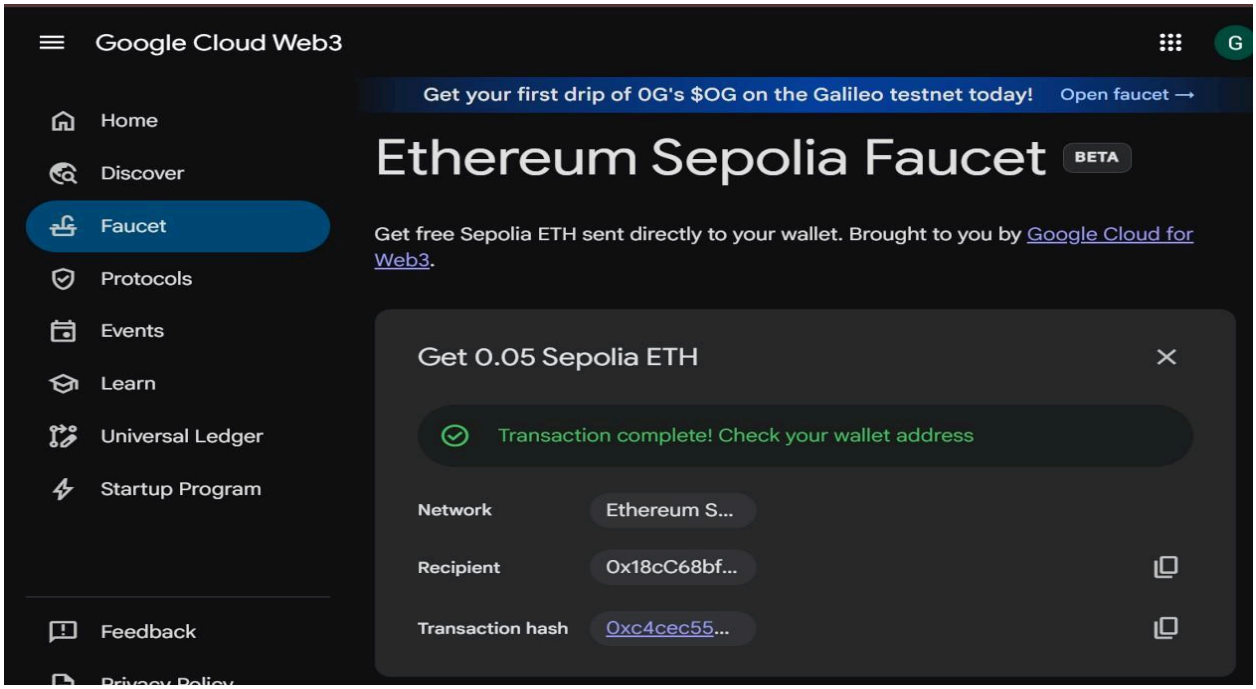


3: In setting, go in Advanced - Turn ON **Show test networks**



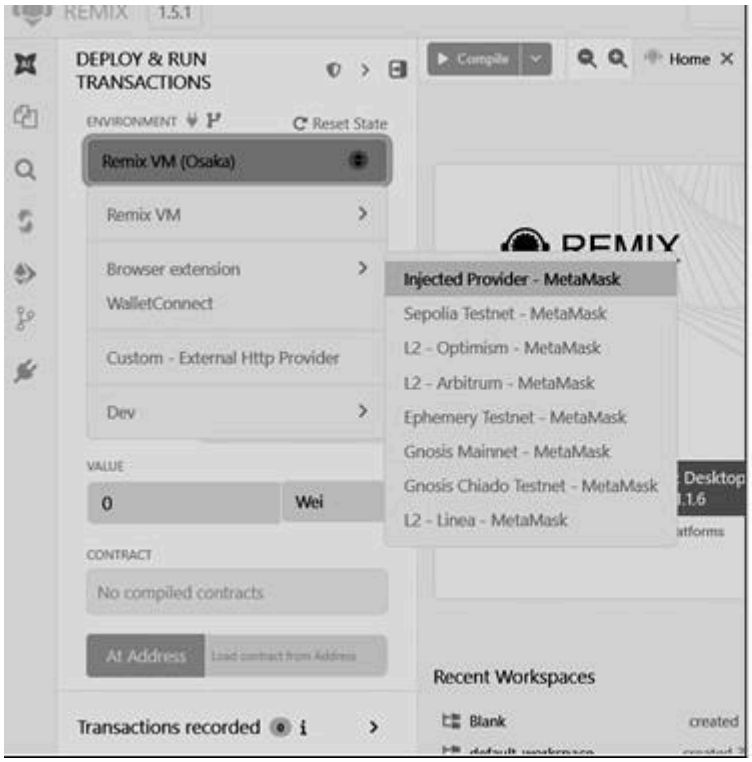


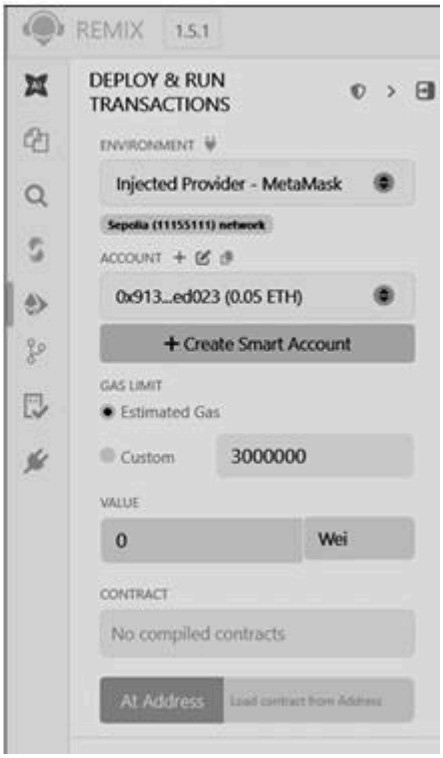
3: Fund Wallet Use faucet to get test ETH



Step4 : Connect the Sepolia Testnet / RSK Testnet to Remix IDE

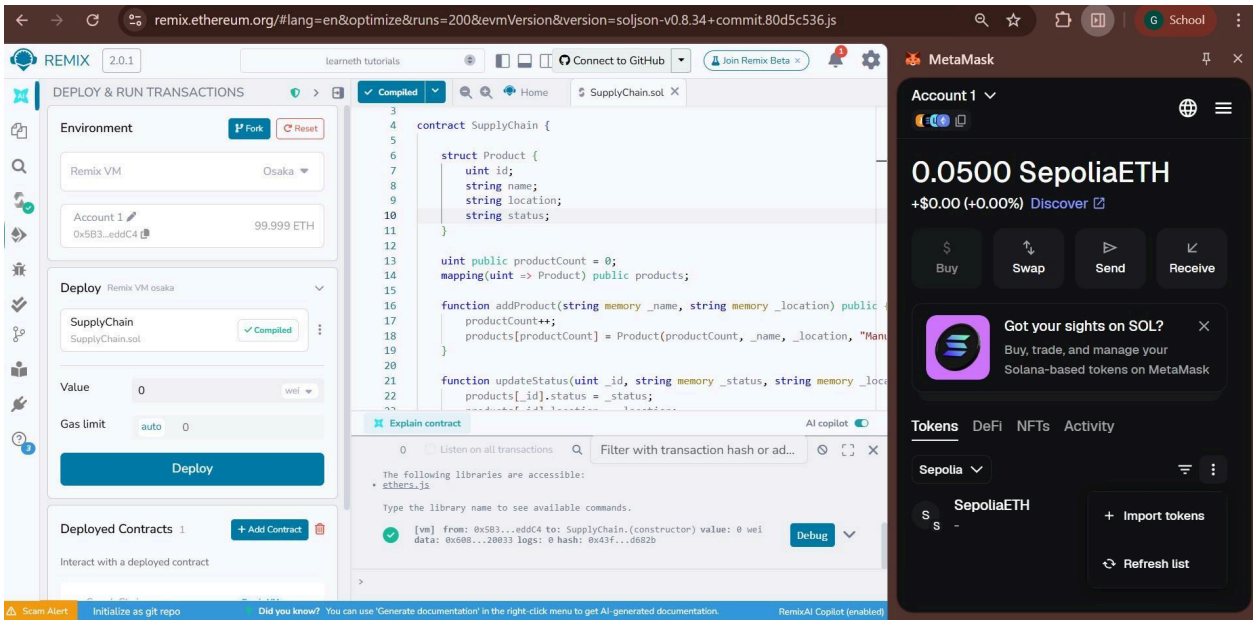
1. In the Remix IDE, select the Environment as Injected Web3.
Note : Injected Web3 connects Remix with the active account in Metamask.





Compile the Contract

1. Open Remix
2. Paste your contract code
3. Go to Solidity Compiler (left panel)
4. Select version 0.8.x (match ^0.8.0)
5. Click Compile **SupplyChain.sol**

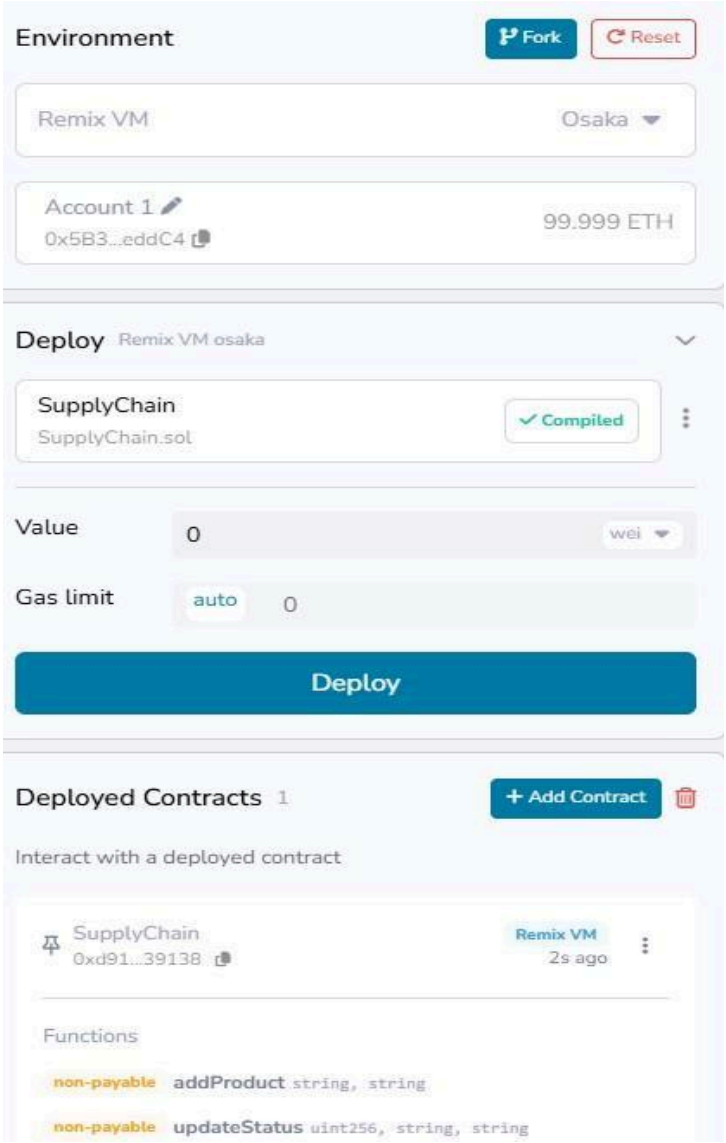


Connect MetaMask (Injected Provider)

1. Go to Deploy & Run Transactions tab
2. In Environment → select Injected Provider - MetaMask
3. MetaMask popup will open → click Connect

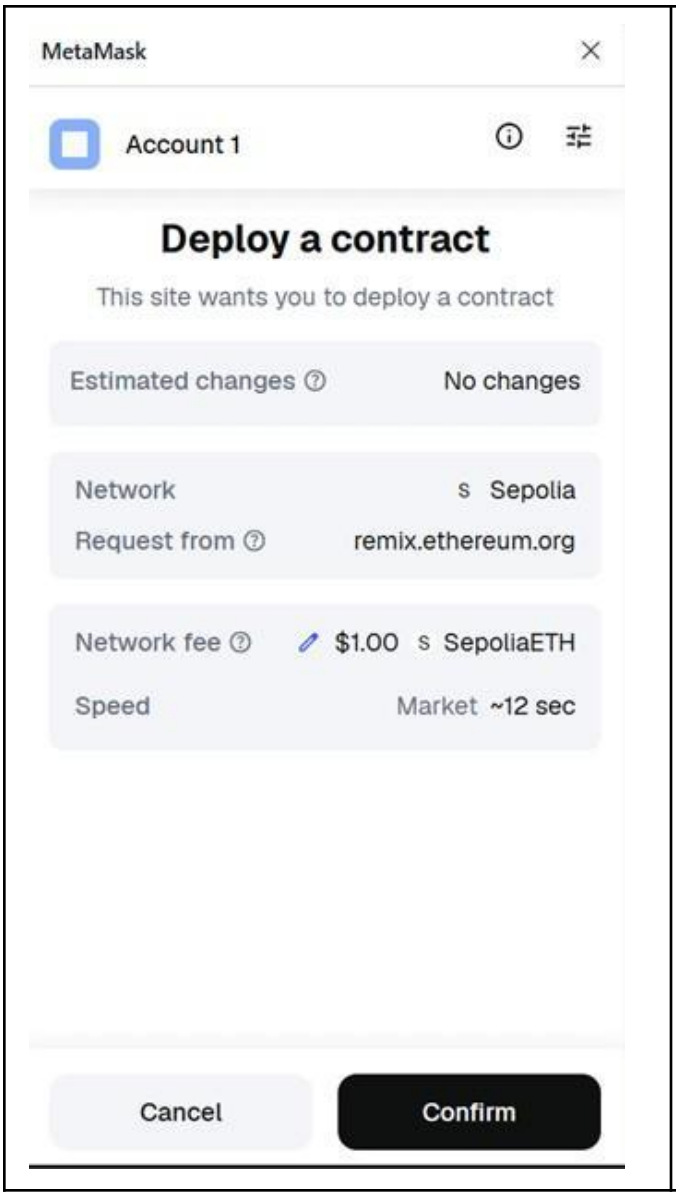
Make sure:

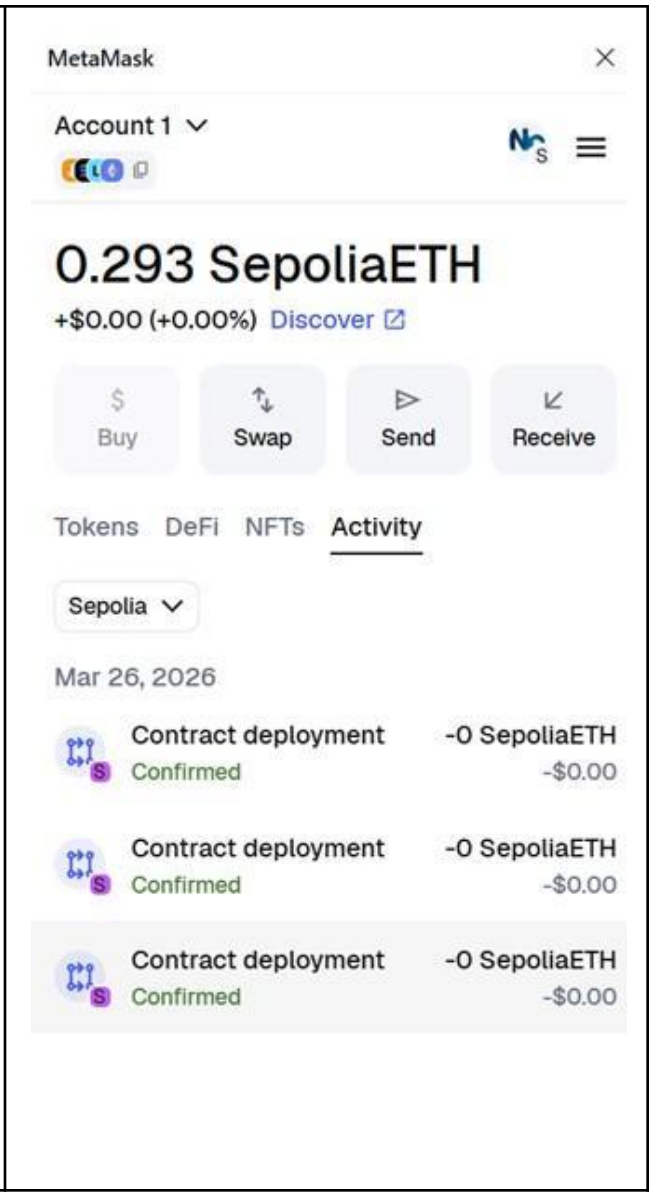
- Network = Sepolia Test Network
- You have 0.05 Sepolia ETH



Deploy Contract
Your constructor needs a supply chain address

Steps:
1. Copy your second MetaMask account address (pharmacy) ○ MetaMask → Account 2 → Copy address
2. Paste that address in constructor field
3. Click Deploy
4. MetaMask will pop up → Confirm transaction
Wait a few seconds…
After success → your contract appears under Deployed Contracts


MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
remix.ethereum.org
Network fee
$1.00 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


MetaMask
Account 1
0.293 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Tokens
DeFi
NFTs
Activity
Sepolia
Mar 26, 2026
Contract deployment
Confirmed
-0 SepoliaETH
-$0.00
Contract deployment
Confirmed
-0 SepoliaETH
-$0.00
Contract deployment
Confirmed
-0 SepoliaETH
-$0.00

Conclusion:

This experiment demonstrates the use of **MetaMask, Solidity, and smart contracts** to build a blockchain-based supply chain system.

By combining:

- **SupplyChainToken** for payment,
- **SupplyChainEscrow** for secure transactions, and
- **SupplyChainRegistry** for record management,

the system provides a practical model for improving **transparency, traceability, trust, and automation** in supply chain operations.

Thus, blockchain technology proves to be an effective solution for modern supply chain challenges.